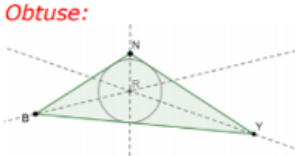
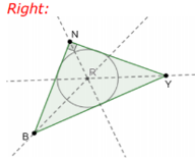
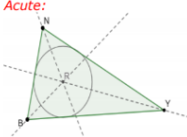


11.2 Constructing the Incenter of a Triangle

Lesson Introduction

 Benchmark Focus	MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle.		 Learning Targets	- I can construct an inscribed circle in a triangle. - I can locate the incenter of a triangle.	
 Benchmark Coherence	Building On			Working Toward	
	N/A			N/A	
 Essential Questions	- How do you construct an inscribed circle in a triangle? - How do you locate the incenter of a triangle?				
 Lesson Narrative	In this lesson, students discover that the point of concurrency of the angle bisectors is the incenter. Then, they construct perpendicular line segments from the incenter to each side of the triangle. Finally, students construct circles inscribed in triangles and compare the locations of the incenter in acute, right, and obtuse triangles.				
 Component Summary	11.2.1 Warm-Up (~5 min.)	Notice and Wonder about triangular pyramid salt piles (<i>SE p. 182</i>)	11.2.2 Exploration (35-40 min.)	Constructing an inscribed circle exploration (<i>SE p. 183</i>)	
	11.2.3 Wrap-Up (~5 min.)	Reflect on most difficult or confusing part of the lesson (<i>SE p. 186</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Incenter <u>Additional Glossary:</u> - Angle Bisector - Point of Concurrency		- Compass - Patty Paper - Straightedge		- Prior to the lesson, work through 11.2.2 Exploration to ensure that you are familiar with the steps students will take and are prepared to support them as they work. - Consider creating, discussing, and displaying a Frayer Model to help students distinguish between the terms “incenter” and “circumcenter.”

 Teacher Tips	Common Misconceptions - Students may not understand how to construct angle bisectors. - Students may not understand how and when to construct an incenter versus a circumcenter. - Students may not recognize that in an equilateral triangle, the incenter and the circumcenter are the same point.	Things to Consider - In 11.2.2 Exploration, consider modeling the constructions and then having students practice independently a few times as they develop fluency with skills. - Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the location of the incenter in right, acute, and obtuse triangles.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In 11.2.1 Warm-Up, students are shown photographs of salt poured onto a triangle until it piles up. Ask students “What do you notice? What do you wonder?” Give students a few minutes to write down things they notice and things they wonder. After students have had a chance to write down their responses, asks several students to share things they noticed and wondered. Record these for all to see. Steer the conversation to wondering about the location of the peak of the salt piles.	MA.K12.MTR.1.1: Participation in Effortful Learning 11.2.2 Exploration is a long and challenging activity. Foster perseverance in students by modeling the steps needed to perform the constructions and recognizing students’ efforts during the activity. Develop students’ abilities to analyze and solve the problems by using the think aloud strategy to demonstrate expectations for student responses to the problems.
 Differentiate Instruction	In 11.2.2 Exploration, students are provided with a kinesthetic entry point as they use patty paper, compasses, and straightedges to construct angle bisectors, an incenter, and a circle. Students are also provided with an auditory entry point as they discuss similarities and differences among the right, obtuse, and acute inscribed triangles. Provide a visual entry point by showing students images and definitions of obtuse, acute, and right triangles. 	
Lesson Notes:		

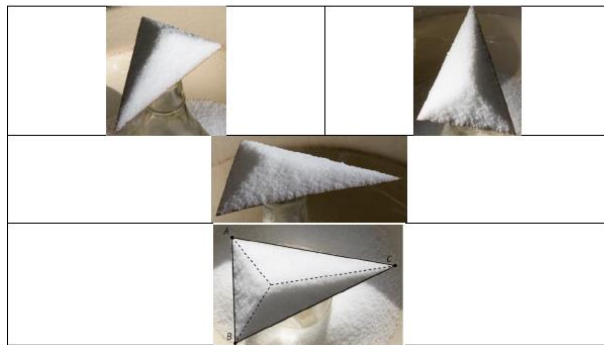
11.2.1 Warm-Up: Salt Pile

Component Overview: Students share what they notice and wonder about photographs of triangular pyramid salt piles.



11.2.1 Warm-Up: Salt Pile

Salt is poured onto a triangle until it piles up. As the salt piles up, the pile reaches a maximum height. As the salt reaches a maximum height, new grains of salt fall off whichever side of the triangle is closest.



Students are shown photographs of salt poured onto a triangle until it piles up. Ask students “What do you notice? What do you wonder?” Give students a few minutes to write down things they notice and things they wonder. After students have had a chance to write down their responses, asks several students to share things they noticed and wondered. Record these for all to see. Steer the conversation to wondering about the location of the peak of the salt piles.

- What do you wonder about the salt pile?
Answers will vary. I wonder how much salt is on the triangle. I wonder how much salt has fallen off the triangle.
- What do you notice about the salt pile?
Answers will vary. I notice that the salt has a point higher than all the others. I notice there is less salt that has fallen off on the back side (AB).
- What do you notice about the point where the salt reaches a maximum?
Answers will vary. I notice that there is more salt at the maximum point than the rest of the triangle. I notice that the maximum point seems to be the same distance from each of the sides.

11.2.2 Exploration: Constructing an Inscribed Circle

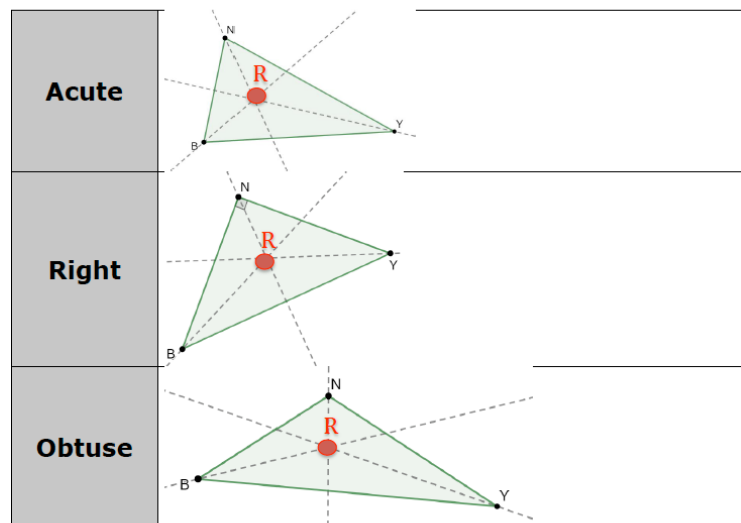
Component Overview: Students discover that the point of concurrency of the angle bisectors is the incenter of the triangle. Then, they construct perpendicular line segments from the incenter to each side of the triangle. Finally, they construct circles inscribed in triangle and compare the locations of the incenters in acute, right, and obtuse triangles.



11.2.2 Exploration: Constructing an Inscribed Circle

1. $\triangle NYB$ is shown as three different types of triangles.

a. Label each point of concurrency as point R .



b. What do you notice about the three different triangles and point R , the point of concurrency?

Answers will vary. I notice that the point R is inside each of the triangles.



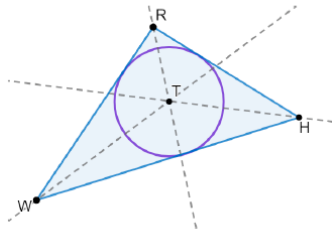
Consider modeling the constructions and then having students practice independently a few times as they develop fluency with skills.



- What do you notice about the point of concurrency of the angle bisectors in right, obtuse, and acute triangles?
- How does this compare to the point of concurrency of the perpendicular bisectors, or circumcenter, in right, obtuse, and acute triangles explored in Lesson 1?

11.2.2 Exploration: Constructing an Inscribed Circle (continued)

2. $\triangle RHW$ is shown where each angle is bisected. The three angle bisectors meet at point T .

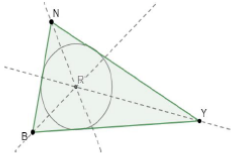


How is circle T related to $\triangle RHW$?

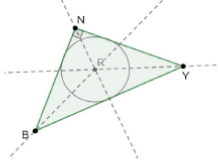
The center of circle T is the point of concurrency of the angle bisectors. The circle is completely inside the triangle.

3. Use a compass to draw the inscribed circles for each type of $\triangle NYB$ in problem 1.

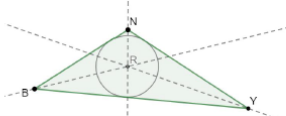
Acute:



Right:



Obtuse:



The point of concurrency of the angle bisectors of a triangle is the **incenter**.



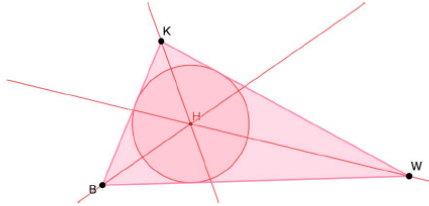
Encourage students to use precise mathematical terms in their written descriptions and verbal discourse. For example, in question 2, highlight students who refer to point T as the “point of concurrency” of the “angle bisectors.” Displaying relevant key terms and an additional glossary on a classroom Word Wall will help students incorporate the new terms into their vocabulary.



Students are provided with a kinesthetic entry point as they use patty paper, compasses, and straightedges to construct angle bisectors, an incenter, and a circle. Students are also provided with an auditory entry point as they discuss similarities and differences among the right, obtuse, and acute inscribed triangles. Provide a visual entry point by showing students images and definitions of obtuse, acute, and right triangles.

11.2.2 Exploration: Constructing an Inscribed Circle (continued)

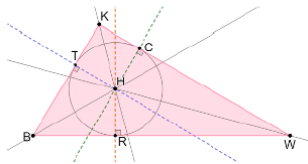
4. Consider $\triangle KWB$.



- a. Copy $\triangle KWB$ onto patty paper and find the incenter of the triangle by constructing the angle bisectors. Label the incenter of $\triangle KWB$ as point H .
Shown above.

- b. Draw the inscribed circle for $\triangle KWB$ using point H , the incenter, as the center of the circle.
Shown above.

- c. Noah is concerned that when he draws the inscribed circle, he does not have the exact points on the triangle where the circle intersects the triangle. He drew three perpendicular lines from the center to each side.



When Circle H is drawn on Noah's version of $\triangle KWB$, what relationship do $\overline{HR}$, $\overline{HT}$, and $\overline{HC}$ have to each other and to Circle H ?

They are the same length because each segment represents a radius of the circle.

- d. Draw Circle H that passes through points T , C , and R . *Students will draw it on the triangle above.*



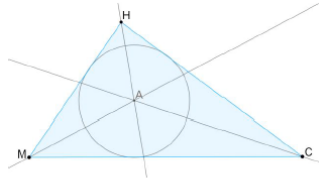
Consider creating, discussing, and displaying a Frayer Model to help students distinguish between the terms "incenter" and "circumcenter."



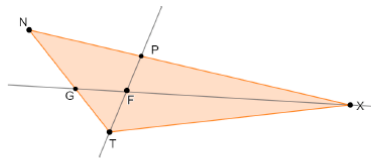
Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the location of the incenter in right, acute, and obtuse triangles.

11.2.2 Exploration: Constructing an Inscribed Circle (continued)

5. Copy $\triangle MHC$ onto another piece of paper. Then, use a compass and a straight edge to inscribe a circle in $\triangle MHC$.



6. $\triangle NXT$ is shown where $\overline{PT}$ bisects $\angle NTX$ and $\overline{GX}$ bisects $\angle PXT$.



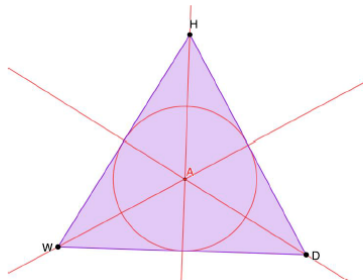
- Will the inscribed circle of $\triangle NXT$ pass through points
- P and G ?

No

- Justify why $\overline{PF}$ and $\overline{FG}$ are not radii of the inscribed Circle F.

The distance from the center of the circle to the sides of the triangle is the perpendicular distance. $\overline{PF}$ and $\overline{FG}$ are not perpendicular to $\overline{NX}$ and $\overline{NT}$, respectively.

7. Copy $\triangle HDW$ onto a piece of patty paper. Then, construct the incenter and circumcenter of $\triangle HDW$.



This is a long and challenging activity. Foster perseverance in students by modeling the steps needed to perform the constructions and recognizing students' efforts during the activity. Develop students' abilities to analyze and solve the problems by using the think aloud strategy to demonstrate expectations for student responses to the problems.



Consider using question 5 as a formative assessment of student mastery of the lesson learning targets. Consider working one-on-one or in a small group with students who are struggling.



- Is it necessary to draw all three angle bisectors to locate the incenter? Why or why not?*
- Is it necessary to construct perpendicular lines from the incenter to each side of the triangle? Why or why not?*



Prompt students to review their work from Lesson 1 to help them construct the circumcenter. Consider having students use different colored writing utensils to construct the incenter than they use to construct the circumcenter. Highlight students' responses as they notice that the incenter and circumcenter are the same point in equilateral triangles.

11.2.3 Wrap-Up: Reflection

Component Overview: Students reflect on the part of the lesson that they found the most difficult or confusing.



11.2.3 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on constructing a circle inscribed in a triangle?

Answers will vary.



This information provided from the students here, coupled with what was observed during the exploration, can provide information for differentiating instruction in later lessons.